Integrating equity in transportation scenario planning: A systematic review

Meiyu (Melrose) Pan, Ph.D.
Oak Ridge National Laboratory
2360 Cherahala Blvd, Knoxville, TN 37932
panm@ornl.gov

Stephen Wong, Ph.D.
Department of Civil and Environmental Engineering
University of Alberta
116 St NW, Edmonton, AB T6G 2E1, Canada
sdwong1@ualberta.ca

Francis Tainter, Ph.D.
Department of Civil and Environmental Engineering
University of Massachusetts, Amherst
131 County Circle, 325 Hampshire House
Amherst, MA 01003, USA
ftainter@umass.edu

Steve Woelfel
Massachusetts Department of Transportation
Office of Transportation Planning
Suite 4150
Boston, MA 02116, USA
steve.woelfel@state.ma.us

Alyssa Ryan, Ph.D.
Department of Civil and Architectural Engineering and Mechanics
The University of Arizona
1209 E 2nd St., Tucson, AZ 85721, USA
alyssaryan@arizona.edu

Abstract

To address the growing uncertainties in the future transportation industry, planners and engineers have identified and implemented scenario planning, a multi-step and strategic approach, in long-range transportation planning. A comprehensive examination of how equity is envisioned, planned, and analyzed during this process remains absent, despite several key guidelines of scenario planning exercises. This paper conducts a systematic literature review focusing on how equity has been considered and evaluated in scenario planning. The resources include academic papers, agency reports, and materials from the recent scenario planning conference, the 2nd Conference on Scenario Planning in Transportation hosted by the Transportation Research Board. This paper examines equity in scenario planning in three important ways: 1) the ways equity has been integrated across steps; 2) how equity analysis has evolved; and 3) when it is necessary to consider equity. This work adds value to the existing body of knowledge by presenting the policy implications based on the review of the current scenario planning research and practices. Early policy recommendations include the need for more vertical equity evaluations, the construction of robustness and justice indicators, and the adoption of a flexible mindset of planning for rather than being fixated on a single scenario.

Keywords: scenario planning; equity; systematic literature review; strategic planning; transportation

1. Introduction

Providing accessible mobility to all members of society, especially those with limited mobility options, is a key responsibility in transportation planning. The transportation sector is confronted with more and more challenges that have an impact on the environment, society, and people. These challenges might arise from significant domestic or global developments such as climate change, emerging technologies, economic factors (e.g., rising fuel prices), or pandemics. The discrepancies in the distribution of the transportation system may be exacerbated by these issues. As a result, long-range planning is crucial for transportation agencies to effectively make decisions that help everyone overcome these future challenges (Brodie 2015).

The complexity of built environments, demographics, behavior, and uncertainty requires careful and thoughtful long-range transportation plans. To aid this planning, agencies have started to utilize scenario planning, a multi-step strategic planning process, to address these issues in recent years (John A. Volpe National Transportation Systems Center 2011). The scenario planning method reshapes transportation planners' traditional mindset about forecasting the future through generating, assessing, and planning for a variety of possible futures (Ravetz, Neuvonen, and Mäntysalo 2021). Due to this method's value in analyzing complicated and uncertain environments, it has been employed across a range of problems and domains including natural hazard mitigation and organizational transformation (Chermack 2011; Moore et al. 2013).

Despite the considerable efforts made by transportation planners to ensure the equitable distribution of transportation resources, inequities have been observed in transportation systems (Brenman 2007; Guo et al. 2020). For instance, the average commute time for late-shift workers, who tend to be low-income people of color and depend on public transit, is twice as long as the commute time for workers who have access to a car (Hughes-Cromwick 2019). Related to gender disparities, women are more prone to perceive public transit as unsafe, uncomfortable, and inaccessible (LA Metro 2019; Pan and Ryan 2023). Other studies have systematically examined the inequitable distribution of active and shared transportation costs and benefits (Lee, Sener, and Jones 2017; Shaheen et al. 2017). These examples and ongoing systemic inequity can be better addressed through a transformative vision for justice in transportation. Progress is quickening as the United States Department of Transportation (US DOT) recently highlighted equity as one of the Department's highest priorities (US Department of Transportation 2022). However, practical and implementable mechanisms for embedding equity in long-range plans remain in their infancy.

While many guidebooks have been developed in regard to general scenario planning approaches, little has been researched on how equity-related factors have been envisioned, planned, and evaluated in scenario planning (John A. Volpe National Transportation Systems Center 2011; Twaddell et al. 2016). Without clear guidelines on equity integration approaches in scenario planning, transportation agencies cannot easily understand where to incorporate equity in their current scenario planning processes and/or where they can further improve their equity analysis methods.

In this research, we conducted a systematic literature review of the existing equity-oriented practices of scenario planning. We note that existing literature already exists on scenario planning in transportation more generally, which can be reviewed in (Chakraborty & McMillan, 2015; Lyons et al., 2021; Sadatsafavi et al., 2019; Zegras et al., 2004). Rather than re-conducting that work, this review paper focused on how and where equity-related factors have been taken into account in scenario planning studies for transportation. Methods regarding how equity analyses in scenario planning have evolved were then summarized. Finally, comparisons were performed between studies that did or did not consider equity.

This paper is organized into four sections. The following section describes the materials and analysis methods for conducting the systematic literature review. The study findings, policy implications, and limitations are then presented. The paper concludes with the study overview with primary research results.

2. Aim and scope of this review

The Federal Highway Administration (FHWA) defines scenario planning as a method of strategic planning that employs alternative narratives of possible futures (or future states) to simulate decisions to make more informed decisions and develop future plans (Federal Highway Administration 2015). Incorporating scenario planning into the creation of long-range transportation plans began as pilot programs in the early 2000s (John A. Volpe National Transportation Systems Center 2011). Since then, FHWA has also created a number of workshops and released several guidebooks in an effort to spread the best practices and methods for scenario planning. Table 1 summarizes the guidebooks that have been sponsored by FHWA.

Table 1. Guidebooks released by FHWA

Title	Year
Federal Highway Administration Scenario Planning Guidebook	2011
Advancing Transportation Systems Management and Operations Through Scenario Planning	2015
Supporting Performance-Based Planning and Programming through Scenario Planning	2016
Next Generation Scenario Planning: A Transportation Practitioner's Guide	2017

The following list from FHWA provides a summary of the steps involved in the general scenario planning process (Ange et al. 2017).

- Phase 1: Define the scope, effort, missions, and goals.
- Phase 2: Create a baseline and alternative scenarios.
- Phase 3: Evaluate scenario impact.

- Phase 4: Select strategic plans and policies based on the analysis.

In more recent scenario planning practices, in some cases, the emphasis in transportation planning has switched from mobility—the ease with which one can use the transportation system—to accessibility—the system's capacity to enable access for all community members (Sanchez 2019). However, there is little documentation in terms of guidelines pertaining to the equity planning approaches throughout the scenario planning processes, as equity-related research in transportation has only recently gained more attention (US Department of Transportation 2022) despite decades of significant research (Bullard 2000, 2007; Foster and Cole 2000; Sanchez, Stolz, and Ma 2003). To better prepare agencies as they embark on scenario planning exercises, this review paper highlights successful equity integration along with missed opportunities.

This paper reviewed the materials from academic research studies and agency reports. Uniquely, this review also brings in discussions from the Transportation Research Board (TRB) 2nd Conference on Scenario Planning in Transportation held in September 2022 surrounding the application of an equity lens in scenario planning. This topic was of primary interest for numerous agencies, including the Massachusetts Department of Transportation, the Louisiana Department of Transportation and Development, and the Central Florida Regional Planning Council (Transportation Research Board 2022). As a result, the presentation and discussion materials from the Transportation Research Board 2nd Conference on Scenario Planning in Transportation were also used as resources for the study. This inclusion allowed for a more comprehensive and timely view of scenario planning practices at state agencies, which are often not documented in research studies or agency project reports.

2.1 Defining transportation equity

Since the 1960s, there has been interests in the distributional consequences of transportation resources, with an emphasis on fairness in the allocation of benefits related to mobility or accessibility as well as adverse environmental and other costs (Linovski, Baker, and Manaugh 2018). In general, the term “transportation equity” is unspecific and can be further defined based on how the benefits and costs should be distributed and who the target audience is (Pereira and Karner 2021). Many different types of theories and working definitions have been developed to understand and assess transportation equity systematically (Karner et al. 2020; Lewis, MacKenzie, and Kaminsky 2021; Rowangould, Karner, and London 2016). Most critically, equity is generally considered a normative concept, which is how the world ought to be (Lewis et al. 2021). We note that Lewis et al. (2021) provide a comprehensive overview of equity in transportation with specific examples of equity theories within the transportation field.

This study focused on the general concept of equity when using the term "equity," rather than more particular definitions, because there was a paucity of scenario planning literature that mentioned equity. The scenario development methods were synthesized from the literature that considered equity. When reviewing the various methods for evaluating the equity impact, this concept was further elaborated by differentiating horizontal equity and vertical equity (Bills and Walker 2017; Litman 2022). Horizontal equity refers to benefits and costs being distributed equally across all groups or spaces. Horizontal equity can be measured by determining transportation service levels, such as frequency and the number of transit routes, their relation to population or employment levels and the spatial distribution of the transit system (Golub and Martens 2014; Wei et al. 2017). On the other hand, vertical equity assumes that disadvantaged groups should receive a greater share of resources. For example, vertical equity justifies planning that favors higher value trips (e.g., healthcare access), more vulnerable travelers (e.g., people with disabilities) and their modes (Biehl et al. 2019; Houston et al. 2004). While there are many definitions and descriptions of equity, this is the definition and concept that was used to develop this review.

3. Materials and methodology

In this study, a standard reporting procedure, PRISMA, was utilized for reporting the inclusion and exclusion criteria in this systematic literature review (Page et al. 2021) .

The literature search was conducted in four widely popular academic databases: Transport Research International Documentation database, Taylor & Francis database, Web of Science, and Scopus. Three sets of keywords were utilized to conduct the search and a series of "AND" operators were used to connect the two sets of keywords. The first set of keywords was "*scenario planning, future risk, or future assessment.*" These keywords were selected because they are all related to long-range planning. The keywords were restricted to document titles to remove the records that are not relevant to this general subject. The second set of keywords was "*transportation planning or urban planning*" to restrict the paper's scope to pertain to the field of transportation planning.

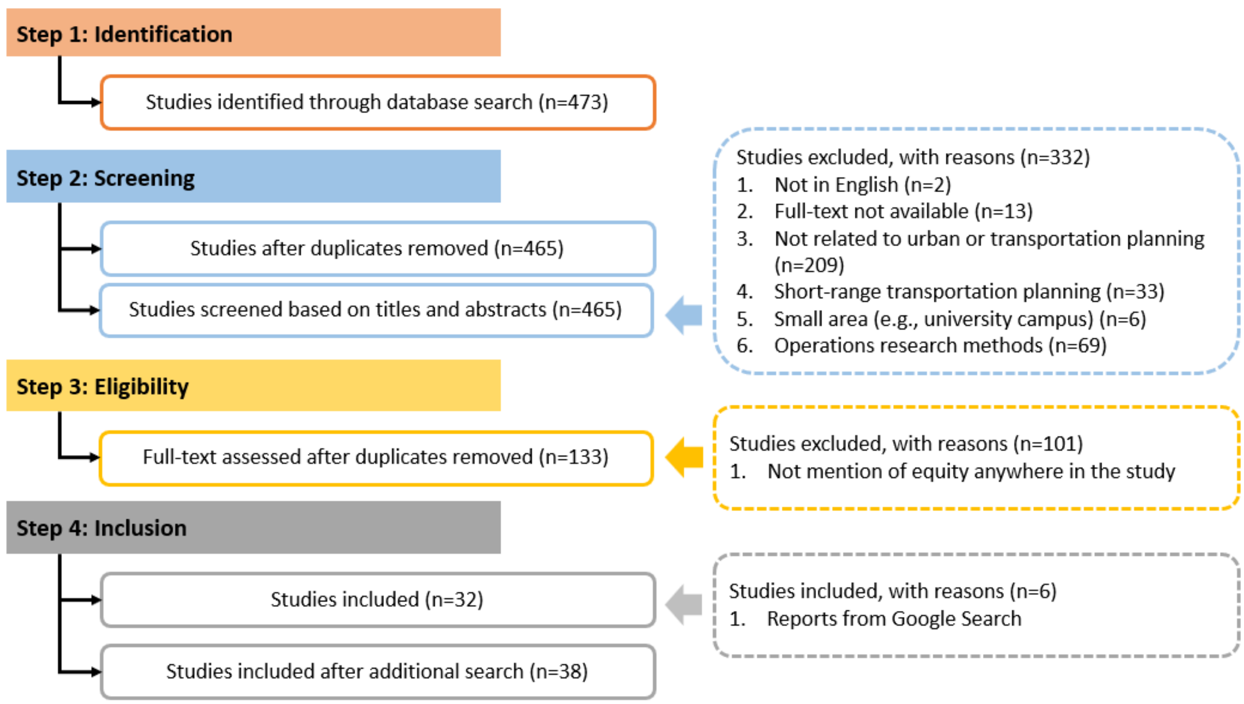

Figure 1. Systematic literature review steps

The inclusion and exclusion process is summarized in Figure 1. The database search identified 473 studies based on the keyword search. In the screening step, studies that were not related to urban or transportation planning (i.e., water systems, biology, and structural engineering) were excluded. After examining the abstracts, studies focusing on short-range transportation planning, small areas (e.g., university campuses), or operations research were excluded. Several non-English, duplicated, and inaccessible documents were also eliminated. This left 133 studies for full-text review. After reviewing the full texts, those without mentioning the concept of equity anywhere in the article were excluded. This review left 32 studies for this paper. This result suggests that the present scenario planning procedures often overlook the concept of equity. Twenty-two of these studies were public reports and nine of them were research publications. Since 2004, the Federal Highway Administration (FHWA) has sponsored 16 scenario planning workshops in 16 states. It is noted that most reports reviewed in this paper are summaries from these workshops.

There were six additional project reports on scenario planning practices that were identified and included. These reports, which were conducted by transportation agencies such as state Departments of Transportation agencies and consulting companies, were collected using a Google search.

To supplement this review, materials and meeting minutes from the Transportation Research Board 2nd Conference on Scenario Planning in Transportation are summarized and integrated into the discussion of the results. This includes an additional 23 presentation slide decks from 16 state Department of Transportation (DOT) agencies and Metropolitan Planning Organizations (MPOs), such as the Massachusetts Department of Transportation and Washington

State Department of Transportation, and seven private companies. All the presentation slide decks are available online (Transportation Research Board 2022).

4. Results

Before presenting the findings, we first start with the definition of a *scenario* used in this paper. A scenario is typically defined as a future world that is influenced by different driving forces, which does not necessarily reflect the agency's proposed policies (S. A. Shaheen, Camel, and Lee 2013). Policies are then evaluated based on each developed scenario. In some practices, however, a scenario is defined as the result of a combination of both driving forces and current policies (Anon 2007). To avoid confusion, in this paper, a scenario is defined as the future world that does not take into account any current or proposed policies. We also note that scenario planning is not equivalent to scenario *analysis* in transportation. This is a more quantitative method that typically involves altering the parameters of models to identify and predict different futures for transportation. Scenario planning is more concerned with qualitative descriptions of possible futures that holistically consider risks that may impact transportation. While its results can be embedded in models, scenario planning is generally used to help guide agencies to consider and respond to future disruptions and alternative futures.

4.1 Which step of scenario planning is the concept of equity considered?

4.1.1 Defining the missions and goals of the scenario

The formation of scenarios depends on the identified goals and missions of the performing agency. It is recommended that agencies further consider the development of evaluation metrics concurrently so that they can reflect these goals (Michigan Dept. of Transportation and Tri-county Regional Planning Commission 2010). For instance, the Delaware Valley Regional Planning Commission (DVRPC) created twelve different metrics for the 2035 long-range plan, each of which relates to its goals, such as improving safety (Twaddell et al. 2016). Some studies, however, did not specifically develop evaluation metrics that correspond to their goals. Particularly in cases where an equitable transportation system was one of the missions, goals, and/or guiding principles, the policy impact on the equitable distribution of transportation system was infrequently evaluated (John A. Volpe National Transportation Systems Center 2011; Lee et al. 2015). The analysis of the distribution of benefits and costs by demographic groups, such as low-income populations, was not particularly included when establishing the equity indicators (Michigan Dept. of Transportation and Tri-county Regional Planning Commission 2010).

4.1.2 Creating scenarios

The creation of scenarios, which is a crucial step in the scenario planning process, constitutes the second step of the process where the concept of equity was often taken into account. Future scenarios (typically four) are developed by intersecting various future trends, also known as driving forces, to create a matrix. These driving forces are usually related to problems with the uncertainty that are outside of the control of a region or agency (Sadatsafavi et al. 2019). Forces are often designed to be extreme to include lower probability but high consequence futures. Among the common driving forces considered are natural disasters (i.e., droughts, wildfires, and flooding), climate change, demographic change, economic change, technological innovation, political change, energy costs, and funding for transportation (Ange et al. 2017). One of the driving forces may be the disparities in the allocation of transportation resources. For instance, factors pertaining to demographic changes, such as aging populations, were also considered in some studies (Neiner, Howze, and Greaney 2004). Once the driving forces are identified, the most impactful and uncertain forces are generally chosen and placed along an axis, which emphasizes the two most extreme directions in which these forces might represent.

The outcomes of the scenario composition process can also be determined by the type of participatory discussions. The discussion usually takes the form of three different types: 1) small-scale within-agency discussions, 2) focus groups with experts, and 3) collective inputs from residents.

Small-scale within-agency discussions

Some agencies sought to create a specific set of scenarios that are driven by a limited number of driving forces. Their method for creating scenarios was mostly based on the agency's goals or the staff members' experiences. For example, the Oregon Department of Transportation first recognized several significant issues that might be harming underserved populations as well as other parts of the community. After analyzing the importance of these issues, they then developed several scenarios based on various energy and transportation system price levels (Twaddell et al. 2016). Similarly, the Washington State Department of Transportation developed three scenarios for State Route 167 based primarily on their objective for a more resilient response and technological capability of employing big data collection and analysis (Transportation Research Board 2022).

Focus groups with experts

More commonly, scenarios were created based on focus groups with experts from various industries, such as public health, transportation, and real estate. Discussions with experts can generate ideas from a variety of perspectives, inspire independent thought, and facilitate the gradual construction of scenarios. To create more informed scenarios and policy recommendations, a wider range of topics compared to those from the former are usually covered that is sometimes beyond transportation such as social problems, political issues, economic trends, environmental issues, behavior change, and technology (S. Shaheen, Camel, and Lee 2013). Some research studies employed the Delphi method, a structured communication technique that was developed in the 1960s. This method can be efficiently employed to conduct in-depth expert conversations. With this approach, focus groups are held in rounds, with the results of the driving forces and/or scenarios being compiled and shared with the group at the end of each round (Renzi and Freitas 2015; Shaheen and Wong 2021).

In some research studies, the scenarios were selected also using quantitative techniques. Quantitative techniques, such as the 2 × 2 matrix, are usually complementary to the qualitative methods to help avoid having an excessively high number of scenarios. This method ranks the driving forces into categories of "high" and "low" priority to generate four possible scenarios in a 2 × 2 matrix (Clark et al. 2022; Shaheen and Wong 2021).

Based on the review of the literature, increasing levels of transportation inequity may not always be explicitly considered a driving force. One possible explanation is that equity might not be considered a driving force that is more critical than other external issues, such as the economy, funding, or climate change.

Collective inputs from citizens

To make the planning efforts inclusive, public outreach has also been incorporated into some scenario planning processes. Some past practices have gathered perspectives from people with diverse backgrounds, such as stay-at-home-parents, bicycle advocates, planners, retirees, environmentalists, transit operators, and special-interest groups, in addition to specialists from various professions (Twaddell et al. 2016). In some studies, agencies created a number of scenarios beforehand, then let the public vote on the most important scenarios for study (John A. Volpe National Transportation Systems 2010). During the TRB conference, the North Dakota Department of Transportation presented the four scenarios they developed, Rural Renaissance, Cities and Centers, Ghost Towns, and Smart and Connected. In order to better assess the possibility of each scenario, they gathered feedback from the general public regarding people's preferred mode of transportation and lifestyles that could represent the four scenarios (Transportation Research Board 2022).

4.1.3 Evaluating policies under different scenarios

After creating scenarios, the next step is to evaluate the impact of future policies, projects, and/or development plans for each scenario to identify the most suitable combinations. Agencies can more effectively prioritize their projects and investments by assessing the policy impact.

Previous studies used both qualitative and quantitative approaches in evaluating the policies under different scenarios. The benefits and limitations are summarized in Table 2. This summary can assist agencies in their identification of appropriate methods based on their own goals and circumstances. More details regarding the evaluation approaches are described in the following paragraphs.

Table 2. Summary of the benefits and limitations of the qualitative and quantitative approaches

Evaluation approach	Benefits	Limitations
Qualitative	Useful for stakeholder communications and/or public engagement	Not always accurate when comparing the trade-offs among policies; may not directly help determine the best policies

Evaluation approach	Benefits	Limitations
Quantitative	Precise and in-depth knowledge about how policies affect equity	Limited data availability; not always easy to measure equity, especially vertical equity

Qualitative approaches

Instead of mathematically evaluating the impact of policies, qualitative techniques in policy evaluation refer to the activities where people assess the potential impact of policies in different scenarios based on their experiences or expertise (Kaufman et al. 2020). To make the evaluation more comprehensive, focus groups or workshops with experts are typically employed in qualitative evaluation. For instance, in one study, participants from various stakeholder groups, including public health, transportation, travel, and tourism, as well as epidemiology, participated in workshops to explore and discuss how particular policies, such as catering to wealthy tourists with high-end products, would cause social disparities (Clark et al. 2022). Potential impact and policy implications were created and documented in response to these conversations.

The results of these qualitative studies can be useful for stakeholder communications or public engagement. To help the public comprehend the potential effects of the scenarios, some agencies additionally created simple-to-understand versions of the scenarios and/or interactive tools that allow people to explore different scenarios and associated policies themselves (Bartholomew et al. 2010; Strauss 2016).

Quantitative approaches

Quantitative approaches involve performing numerical analysis to estimate how policies related to developed scenarios would perform. In one study, different equity indicators were created, ranging from sustainability indicators to mobility indicators to land use indicators (Chakraborty 2011). Overall, results from quantitative approaches can provide more precise and in-depth knowledge of how policies affect the equitable distribution of policy impact. An illustration of the methods and analyses employed in scenario planning to evaluate horizontal and vertical equity is presented in Figure 2.

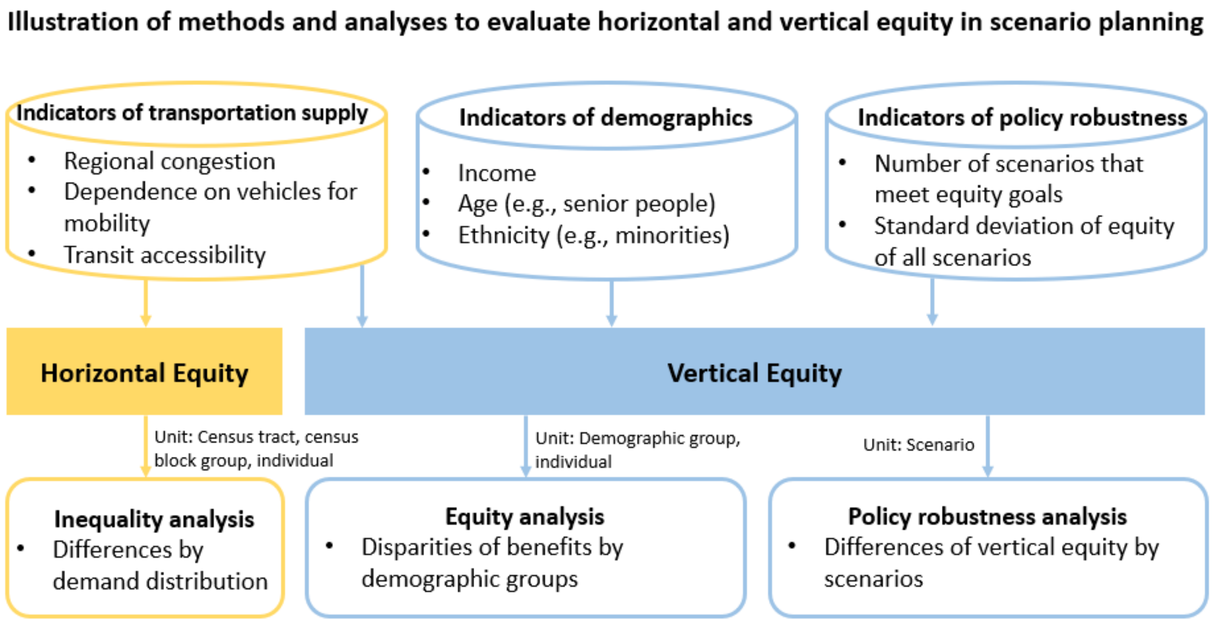

Figure 2. Illustration of the methods and analyses from literature to measure horizontal and vertical equity in scenario planning (adapted from Boisjoly & El-Geneidy's book chapter (Boisjoly and El-Geneidy 2021))

Concerns were also raised about the possibility that equity-related problems and stakeholders' values might not always be embedded in quantitative analysis. The causes of this might be due to an inadequate amount of data and/or a considerable computational burden. As a result, methods for capturing these factors that can affect planning and policy-making should also be developed (Avin 2016).

<u>Horizontal equity indicators</u>

Horizontal equity, also referred to as fairness and equality, concerns how the impacts of a scenario planning process are distributed among individuals with similar needs and capabilities (Litman 2022). It primarily aims for two goals: a fair allocation of resources and an equitable distribution of external costs. Equity analysis typically involves the examination of various outputs, including public facilities and services (e.g., government subsidies), user costs and benefits (e.g., transit fares, road tolls), service quality (e.g., the availability of transportation modes, transit service reliability), external impacts (e.g., congestion and crash risk imposed by an individual or group on other road users), and economic impacts (e.g., access to employment) (Chittenden County Metropolitan Planning Organization and Federal Highway Administration Vermont division 2008; Denver Regional Council of Governments 2013).

However, it's important to note that while scenario planning often incorporates these outputs, they do not directly represent equity impacts. For instance, congestion has frequently been used as one of the performance measures of scenario planning (Ange et al. 2017). However, the relationship between congestion and equity issues is not immediately evident. While some individuals (those who frequently drive during peak hours) bear a greater share of congestion costs, they also contribute to causing congestion. Moreover, automobile traffic delays can affect pedestrians and buses, imposing costs on other road users. Therefore, to thoroughly assess horizontal equity, an additional step of equity analysis is required.

Several equity evaluation techniques applicable to the transportation domain have been summarized and can be further employed in the field. For instance, one such technique involves estimating the theoretical allocation of infrastructure spending in accordance with horizontal equity principles and comparing it to the actual expenditure for a specific user group. For example, one study conducted an equity analysis by assessing the theoretical allocation of funding for non-auto modes and comparing it to the actual spending on various aspects such as parking subsidies, traffic services, roads, paths, and operating subsidies (Litman 2018). The findings revealed that horizontal equity could justify allocating 10% to 30% of infrastructure spending to non-auto modes to ensure that users receive their equitable share. However, upon comparing these findings to the real-world spending, it became apparent that individuals relying on non-auto modes, or those interested in using them, were receiving less than their fair share of investments. Additionally, equity analysis can be focused on external costs, including the delay, risk, and pollution damages imposed by travelers on other individuals. For instance, this analysis can examine congestion delays experienced by users of space-efficient modes such as buses and rideshare vehicles versus those employing space-intensive modes like automobiles (Park et al. 2020). Likewise, it can compare crash risks for pedestrians and bicyclists and automobile users (Bouaoun, Haddak, and Amoros 2015).

Different evaluation practices can result in varied equity assessments. A multi-dimensional approach to evaluating equity is recommended in the literature, such as multimodal analysis, consideration of temporal changes, assessment at different aggregation levels, and accounting for diverse policy objectives (Guo et al. 2020). For instance, an exclusive focus on individual transportation modes can yield differing conclusions. Evaluating equity within each mode separately might reveal that one mode (e.g., public transit) receives more subsidies per user than another (e.g., private car users). However, a system-wide analysis may expose an overall resource distribution that favors car users due to their numerical majority (Basso and Jara-Díaz 2012). Moreover, assessing equity over time can yield divergent findings. Furthermore, assessing equity over time can yield differing results. A single-point assessment may overlook changes in access and service quality over extended periods, while trend analysis provides a more dynamic perspective on equity (Guzman and Hessel 2022). Additionally, the choice of aggregation level for data analysis can impact outcomes. Studies have shown that higher levels of aggregation tend to obscure disparities among individuals (Chen et al. 2019; Pandey, Brelsford, and Seto 2022). Moreover, as discussed in Section 4.1.1, the prioritization of specific policy goals can significantly influence equity assessments. For instance, if a transportation agency places a high emphasis on reducing congestion, resource allocation may disproportionately benefit car users, potentially leading to equity conclusions that favor one group over another. Conversely, if economic benefits take precedence in scenario planning evaluations, the transportation needs of disadvantaged communities might be overlooked (Luo et al. 2022; Tu and Marcantonio 2016).

For horizontal equity examination, census tracts, census block groups, or parcels are typically the units of these indicators (Guthrie, Fan, and Das 2017). Some metrics have further been developed at the individual level. To evaluate several policies, including transit extensions and pricing schemes, one study used individual households as a unit to construct the accessibility and consumer surplus metrics (Bills 2013). Smaller granularity metrics were found to solve the issue of the indicators at the zonal level and average measurements of equity indicators, which are likely to mask the inequitable effects of transportation policies.

When presenting the results of scenario planning to non-technical audiences, higher granularity metrics may be advantageous (John A. Volpe National Transportation Systems 2010). One study showed the final results of the policy's impact on traffic and transit availability in GIS at a large geographic scale. In contrast to being overwhelmed by results at the neighborhood or street level, they claimed that the audience can concentrate on higher-level, regional outcomes due to the display of lower granularity. Focusing on outcomes at a larger scale was also thought to facilitate more productive discussions among stakeholders and the general public regarding the future of the region (Michigan Dept. of Transportation and Tri-county Regional Planning Commission 2010).

To quantitatively assess the impacts of various policies or development plans in scenarios, simulation tools were widely used. This is because assessing the impact of a policy typically requires considering housing, land use, infrastructure, transportation, and the environment at the same time. However, since simulations take too long to generate the outcome for each policy and scenario, simplified versions have been developed to accelerate the evaluation process. Four platforms are most commonly used by agencies for GIS-based scenario planning, three of which fall under the category of sketch planning tools. Simplified tools have been designed to speed up the simulation method because simulation takes a significant amount of time to generate the results for each policy and scenario.

<u>Vertical equity indicators</u>

Many studies have investigated horizontal equity indicators, but relatively few have addressed vertical equity. Vertical equity can be described as the effects of transportation facilities, services, and/or policies on demographic attributes including age, gender, and income (i.e., between groups of differing needs). The lack of research on vertical equity in scenario planning might be because there are not enough demographic data available, particularly for the examination at a finer level (Avin 2016). One study developed vertical equity metrics on person-trips for low- and middle-income cohort evaluation metrics. The results showed that among all developed scenarios, nearly all of them achieved the identified goal of achieving a certain level of emission reduction. Less than half of the scenarios, however, led to the desired outcome of achieving an equitable person-trips reduction (Lempert et al. 2020). This research study suggests that vertical equity evaluation can more accurately reflect a policy's capability, making it valuable for selecting the most effective policies.

Robustness indicators

According to the literature, a robust policy is defined as a policy that can function or work well under a variety of possible futures or scenarios (Capano and Woo 2018; Walker, Rahman, and Cave 2001). One policy change can have different effects on equity indicators, such as transit accessibility, quality of life, public health, and affordability, depending on the scenario. Some policies might be successful in increasing most of the equity indicators in one scenario but not in another.

To overcome this challenge and achieve a balance, agencies may gain a better understanding of the advantages and disadvantages of their identified policies, projects, and/or investments by conducting a robustness analysis. For instance, a policy's worst performance, which represents the most negative impact the policy may have on equity, might be a useful indicator of its robustness. Agencies may prefer a more robust policy if its worst impact is better than that of other policies based on the evaluation results from all scenarios. In one study, the metrics of the overall performance rating for a policy across all scenarios and the proportion of scenarios where the equity goals are achieved were utilized to assess the robustness (Lempert et al. 2020). The robustness of policies relating to equity, however, has not been extensively studied in the present scenario planning research or practices.

Mixed methods

A mixed approach has also been used in the literature. In one study, a group of specialists began the evaluation by identifying potential indicators that might be employed in the policy evaluation process. The number of indicators was then narrowed down based on the data availability and focus group meetings. Based on the discussion outcomes, the agency then utilized a quantitative analysis approach to evaluate the policies in each scenario (Twaddell et al. 2016).

4.2 How has equity analysis evolved?

Although extensively studied, scenario planning as a long-range transportation planning approach has not yet been widely adopted by many agencies. FHWA has led 16 workshops throughout 16 states since 2004. Only 15% of MPOs reported employing a scenario planning approach in a 2013 FHWA survey (Federal Highway Administration 2013). The lack of funds to hire experienced staff or consultants, the workload of the current staff, and the staff's limited expertise in scenario planning were some of the identified challenges. The adoption of scenario planning, according to a workshop in 2017, follows a typical "diffusion of innovation" curve: some agencies are leading the way, many are experimenting with or using scenario planning sparingly, and a few have not considered the method (Snyder 2017).

According to the literature review, equity has been brought up in scenario planning since its start. The trends for equity integration in scenario planning are outlined in Figure 3.

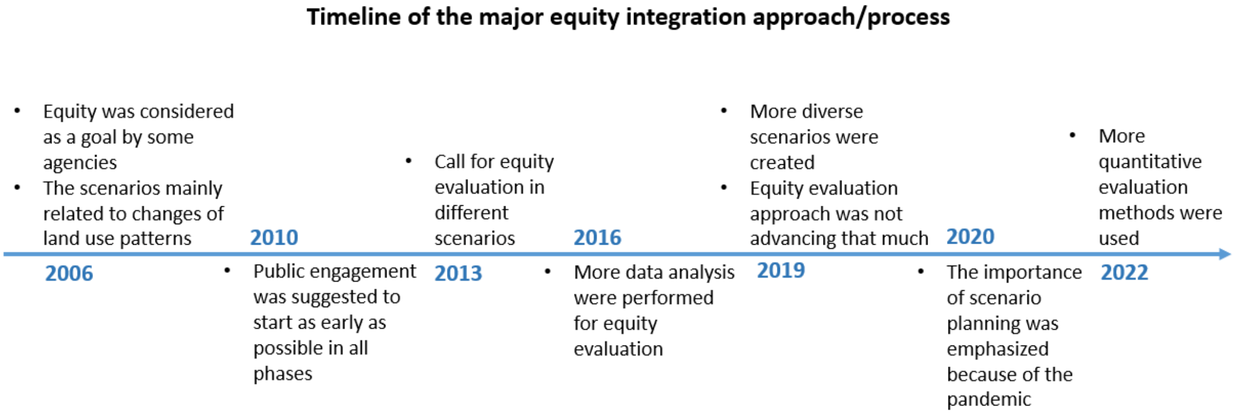

Figure 3. Trend of the major equity integration approach/process

The earliest study that incorporated equity in scenario planning identified by this review paper was published in 2006 when the California Regional Blueprint Planning Program stated that one of its goals was the integration of equity in transportation during their scenario planning workshop (Federal Highway Administration 2006). The scenarios and policies at the time were focused on land use changes, such as transit-oriented cities, larger cities, and smaller cities. Since then, scenario planning has also incorporated more public involvement strategies (Michigan Dept. of Transportation and Tri-county Regional Planning Commission 2010). There were more discussions about scenario evaluation, particularly which equity indicators should be used, in a scenario planning workshop in 2012 (Denver Regional Council of Governments 2013). The majority of evaluations were still qualitative, but some academic studies began to emphasize quantitative methods for equity evaluations. For example, equity metrics at the individual level were investigated, which reduced the biases present in average equity indicators (Bills 2013).

FHWA developed a guidebook in 2016 that focused on the performance assessment of scenario planning. This report thoroughly examined the challenges in implementing equity evaluation into scenario planning, such as the limitations of spatial models and data resources (Twaddell et al. 2016). At that time, scenario planning emphasized the horizontal equity evaluation process. Among the identified metrics was accessibility to public transit, availability of job training, and the general affordability of mobility options. GIS technologies and related qualitative tools were frequently employed for data analysis (Avin 2016; McBrien and John A. Volpe National Transportation Systems Center (U.S.) 2018). After 2019, more scenarios were developed given the emerging challenges and taking into account growing technology (e.g., use of wireless sensors on vehicles and road infrastructure, adoption of driverless-car technology) (Garrick 2019; Higgins 2019). These disruptions to the transportation system sparked more scenario planning conversations. However, equity was not thoroughly examined in these emerging scenario planning exercises.

The need for adopting a scenario planning approach to effectively prepare for future uncertainties in the world has been widely acknowledged recently due to the onset of the

COVID-19 pandemic. Shaheen and Wong (2021) developed a scenario planning exercise for public transit and shared mobility just several months after the start of the pandemic. They divided their exercise in three progressive timeframes, and the workshop participants developed a series of policies that could be grouped in four major categories: 1) immediate policy actions across actors; 2) alignment of societal objectives; 3) federal transportation spending reauthorization; and 4) finances and subsidies. Key equity-oriented policies surrounded a traveler's bill of rights, mobility-for-all strategies, targeted service for underserved populations, and new metrics for assessing equity. The research was also presented as a policy brief for improved implementation (Shaheen and Wong 2021) and placed in the wider context of the pandemic's impacts on transportation (Shaheen and Wong 2023).

Similarly, at TRB's 2nd Conference on Scenario Planning in Transportation, multiple DOTs and MPOs presented their scenario planning procedures, equity evaluation techniques, and lessons learned from the 2022 COVID-19 pandemic. Discussions focused on how the pandemic affected minority, low-income, and essential employees in particular. Additionally, a wider variety and more innovative scenario planning techniques and community participatory approaches, including models and focus group research, have been developed and adapted in recent years (Lyons et al. 2021; Peregrino et al. 2021). One important lesson learned from the conference was that transportation agencies can respond to future risks more rapidly and effectively if they had adopted a mindset considering numerous futures rather than just one. With this mindset shift, agencies were able to more actively face uncertainties with flexibility and nimbleness, as opposed to just ignoring or avoiding challenges (Page et al. 2010; Rohr et al. 2016). Applying an equity lens throughout scenario planning can further lead to more robust policies (Chakraborty et al. 2011; Shaheen and Wong 2021).

4.3 Circumstances of equity consideration

Even though an equitable distribution of transportation resources is a vital goal in transportation planning, many other considerations could take precedence. Four examples of research studies were examined and compared to understand when equity aspects should be integrated. Two of the four studies did not specifically take equity into account, whereas the other two did. The two sets of studies were selected based on the fact that they employed the same driving forces that are prevalent in recent scenario planning practices. Two studies developed scenarios based on different levels of transportation funds and economy for regional transportation planning. The other two studies, which focused on tourism planning, developed scenarios based on the uncertainties of tourism's supply and demand. The summary of the studies is provided in Table 3.

Table 3. Summary of the comparison studies

Author, year	Objective	Driving forces	Equity consideration (Y/N)	Evaluation metrics	Key policy outcomes
(Shaheen and Wong 2021)	Regional transportation planning	Economy and transportation funds	Y	1) travel experience 2) effectiveness of public transit	1) customer experience 2) equitable planning and workforce development
(Sadatsafavi et al. 2019)			N	1) land use type 2) economy	1) multimodal transportation 2) innovative technology
(Clark et al. 2022)	Tourism development	Mobility trend and transportation supply	Y	1) demographics 2) public health	1) affordable recreation 2) health and safety protocols
(Page et al. 2010)			N	1) economics 2) sustainability	1) connectivity 2) sustainable mobility option

Based on these four examples, it appears that equity considerations were more commonly observed when the experiences of transportation system users played a significant role in scenario planning. Conversely, studies that did not consider equity tended to emphasize system efficiency, economic growth, and/or technological aspects.

The identified policy outcomes can also be influenced by whether equity-related issues are considered. According to the comparisons, improved connectivity and accessibility were found in the policy outcomes of all four studies. However, the studies that considered equity emphasized the necessity of advancing the policy benefits for all people. The studies that did not consider equity primarily generated other broader policies, such as the development of new technology and the improvement of environmental mobility options.

5. Discussion

5.1 Overview of findings

In this research, a systematic literature review was conducted to understand the equity considerations in the current scenario planning processes. The resources for this review paper included academic papers, agency reports, and recent scenario planning conference materials. The findings were arranged according to three major sections: (1) how equity was considered in each planning step; (2) how equity analysis has changed through time; and (3) how equity was evaluated in scenario planning.

Equity has been considered primarily in three steps: 1) defining the missions and goals of the scenario, 2) composing scenarios, and 3) evaluating policies under different scenarios. Particularly in the evaluation of policies, equity is crucial. In the literature, qualitative approaches, quantitative approaches, and combined approaches have all been employed. While quantifying equity has not been extensively applied due to constraints (including those in data availability and advanced model forms), the method can provide more thorough and accurate insights into policy selection. When engaging the public and communicating with stakeholders, on the other hand, qualitative evaluation might be more helpful in displaying the big picture.

Based on the review of the history of equity analysis methods in scenario planning, the construction of more diverse scenarios involving new technology and climate change has been observed. However, equity evaluation under these new scenarios has yet been fully examined. More quantitative equity evaluation methods have also been recommended in recent years. We also found that scenario planning with and without equity considerations results in different policy outcomes. The studies that specifically included equity tended to focus on improving the experiences of all users, whereas the studies that did not include equity were inclined to focus on growth strategies.

5.2 Policy recommendations and proposed guidelines

In this section, we developed a series of recommendations and guidelines derived from the results of the review.

5.2.1 Policy implications derived from each results section

The key policy implications regarding each section in Section 4 are discussed in the following paragraphs.

5.2.1.1 Steps of scenario planning

Policy recommendation 1: Equity should be embedded throughout the scenario planning process, ideally at each stage.

As described in Section 4.1, equity can be an unspecific concept when defining the missions and goals of a long-range transportation plan. The definition of equity should be clarified when creating scenarios and evaluating policies. To further this development, major demographic trends (such as aging) could be considered as one of the driving forces when composing scenarios. Quantifiable equity metrics could also be developed to examine the performance of different policies. Even if equity may not always be the top priority for all agencies, taking equity impacts into account in all scenarios may help to encourage the selection of more human-centered policies.

In particular, it is critical in the scenario planning process to conduct a robustness analysis to identify equitable policies that would be successful across all scenarios. As presented in Section 4.1.3, robustness analysis (which focuses on mitigating worst-case effects from a policy) can help guide agencies in the final stages of scenario planning. Through this integration in scenario planning, agencies can make more targeted and equity-centered decisions. In other words, the likelihood of the worst-case situations for underserved populations could be minimized with the use of the robustness analysis.

Policy recommendation 2: Horizonal equity analyses should not just focus on the outputs such as congestion, but also their equity impacts.

While scenario planning often involves an examination of various outputs such as emissions, mobility, and accessibility, they do not directly represent equity impacts. To thoroughly assess horizontal equity, an additional step of equity analysis could be conducted. Table 4 provides a summary of various types of equity analysis that can be performed using scenario planning output.

Table 4. Equity analysis using scenario planning outputs

Typical outputs of scenario planning	Equity impacts that could be analyzed	Comparison groups
Public Facilities and Services	- Infrastructure spending - Road space	- Comparison between demographic groups - Comparison between different modes of transportation - Cost-benefit analysis
User Costs and Benefits	- Road tolls for - Parking costs	- Comparison between users with different VMT per time period - Comparison between demographic groups
Service Quality	- Number of travel modes available in an area (e.g., walking, cycling, delivery service) - Roadway quality (e.g., physical condition) - Public transportation service quality (e.g., frequency, speed, reliability) - Universal design (e.g., accommodation of people with disabilities and other special needs)	- Comparison between different areas/regions, such as urbanites and suburbanites - Comparison between demographic groups
External Impacts	- Congestion delays - Crash risk by multimodal users - Traffic noise and air pollution	- Comparison between demographic groups - Comparison between by users of different transportation modes - Comparison analysis between motorists and non-drivers
Economic Impacts	- Access to education and jobs - Business activity, property values, and economic development in an area	- Comparison between demographic groups - Comparison between motorists and non-drivers - Comparison between different areas/regions

5.2.1.2 Evolution of equity analysis

Policy recommendation 3: Vertical equity analyses should be integrated into the scenario planning process.

As can be discussed in Section 4.2, vertical equity evaluation in scenario planning can benefit from both qualitative and quantitative approaches. To ensure identification or confirmation of all community values, vertical analysis can make the inclusion of a wide participant base possible based on demographic characteristics, socioeconomic factors, and particular issues agencies face. Vertical equity analyses can be applied in a variety of aspects of this process. For example, vertical equity can be analyzed by how costs and benefits should be allocated based on people's mobility needs, income, and/or structural inequalities including racism and sexism (Litman 2022). Additionally, the ability to collect data, such as Census data, regional or local surveys, and in-depth focus groups, can inform vertical equity analyses (Liu et al. 2022).

Policy recommendation 4: Advanced community participation techniques should be utilized to help increase vertical equity principles.

Community participation is needed to further integrate equity into scenario planning procedures. In recent years, more agencies have employed focus groups, interactive exercises, and games at public meetings and workshops in their scenario planning processes, according to the literature and the discussions from the Transportation Research Board (TRB) 2nd Conference on Scenario Planning in Transportation (Transportation Research Board 2022). However, more advanced community participation approaches should be explored. This focus on community participation to build justice-oriented outcomes (Karner et al., 2020) can be furthered by training community leaders and members of local communities about the scenario planning process (e.g., identifying driving forces, developing scenarios) and how to conduct data analysis (e.g., vertical equity analysis methods). In this way, people can understand how they can more effectively convey their suggestions and how their participation can affect long-term plans (Karner et al. 2020). Community participation is also critical to help agency staff have a more comprehensive understanding of environmental justice, cultural competency, and community involvement strategies(Aimen and Morris 2012).

5.2.1.3 Circumstances for equity consideration

Policy recommendation 5: Equity should be analyzed in emerging, more immediate, scenario evaluation.

Equity has not been widely considered in scenario planning practices, especially when the goals focus on the development of the economy and/or technology (see Section 4.3). However, it is still important to consider equity in these emerging, more immediate, and new scenario discussions to ensure an equitable distribution of benefits of costs of the new technologies. For example, studies revealed that higher-income households who make up a larger share of the electric vehicle market receive both a disproportionate amount of government subsidies as well as benefits (Carlton and Sultana 2022; Sheldon 2022). Inadequate infrastructure and/or services in some communities, inequitable incentive designs, and a lack of affordable technology can all result from not addressing these issues during the planning stage for these more immediate issues.

The mindset that scenario planning employs for planning for these uncertainties is essential to form more informed decisions, and to develop more flexible and nimble plans when futures do not turn out as forecasted. One challenge is that the data might not always be available, especially for these novel scenarios. It is worthwhile to begin scenario planning by applying some qualitative techniques or using pre-existing or survey data to find preliminary estimates for scenario generation and/or policy evaluation. As more data is gathered, scenario planning iterations can be undertaken more frequently, including for long-range and more immediate needs, resulting in the development of informed decisions and policies.

5.2.2 Guidelines to improve the scenario planning process

Our results raise an important question: how we can equitably improve the scenario planning process? Based on the key phases of scenario planning, this study proposes a flexible guideline for jurisdictions to integrate equity throughout scenario planning as follows:

- Phase 1: Define the scope, effort, missions, and goals.
    - Consider equity as one of the goals of the long-range transportation plan.
    - Identify a working definition for equity.
    - Discuss how certain populations and groups benefit or are hurt by existing planning processes.
    - Develop potential equity-centered metrics for the evaluation of scenarios and policies.
- Phase 2: Create a baseline and alternative scenarios.
    - Develop a series of potential driving forces based that span wide categories, such as **S**ocial **P**olitical **E**conomic **L**egal/Policy **T**echnological **+** Other (SPELT+).
    - Evaluate if the disparities in the allocation of transportation resources is a critical driving force.
    - Conduct community participation to gather opinions and preferences from people from various perspectives, contexts, and backgrounds.
- Phase 3: Evaluate scenario impact.
    - Conduct community participation to assess the perceived impact of policies in different scenarios, especially with underserved populations.
    - Analyze how each scenario effects equity through different lenses (e.g., accessibility, vertical vs. horizontal, etc.) based on the evaluative metrics developed in Phase 1.
- Phase 4: Select strategic plans and policies based on the analysis.
    - Structure policies such that they focus on the specific needs of underserved populations *and* holistically address equity needs across the jurisdiction.

- Identify equitable policies that would be successful across all scenarios, especially "worst-case scenarios."

5.3 Study limitations

First, the definition of equity is not comprehensive in the literature search phase. This study selected literature based on just the keyword of "equity." Other equity-related concepts such as accessibility or community-centered approaches were not used in the document/literature selection. Secondly, our study primarily focuses on vertical and horizontal equity, but it's important to recognize that various other quantitative methods for equity evaluation were not covered here. As equity becomes further integrated into scenario planning, additional reviews and analyses on equity perspectives, as discussed extensively in (Cabello, Hyland, and Marantz 2023; Lewis et al. 2021), will be necessary. Third, as many states have not yet adopted a scenario planning approach, this review paper primarily focuses on a small number of studies. Furthermore, some scenario planning practice documentation, including the specific steps and lessons learned, was not publicly available. Fourth, the characteristics of an area or state may influence a jurisdiction's decision to analyze and evaluate equity in its scenario planning process. These characteristics may be associated with geography, types of land use, populations, politics, or agencies. These elements, however, were not specifically examined in this research. Future incorporation of the relationships between these elements and equity analysis methods can serve as a comprehensive reference for agencies to identify context-specific approaches. Finally, as the focus of this study is on how the concept of equity was included in scenario planning processes, the specific equity theories or indicators used in the scenario planning literature were not comprehensively synthesized. Since scenario planning exercises were not explicit in their use of an equity theory or framework, we were unable to conduct an analysis.

6. Conclusions

This paper conducts a systematic literature review focusing on equity analysis in the scenario planning processes. The resources for this review paper include academic papers, agency reports, and recent scenario planning conference materials. As part of its contribution to research and practice in scenario planning, we specifically examined where equity was considered, how equity analysis has evolved, and when it is necessary to consider equity. The key findings from the literature review are summarized in the following bullet points.

- Scenario planning for future policy development must take equity into account if systems are to provide transportation for all users. Qualitative evaluation can be helpful during stakeholder communication and public engagement, while quantitative evaluation can provide more thorough and accurate insights into policy selection.
- In recent years, more diverse scenarios involving advanced technology have been constructed. However, a thorough examination of equity evaluation in these novel

scenarios has been largely absent. Another notable trend pertains to the increased use of quantitative methods for policy evaluation in scenario planning.

- The practices that consider equity tend to focus on enhancing the experience of all users, whereas the studies that do not specifically consider equity tend to concentrate on the economic and/or technological advancement of the transportation sector.

We also contribute to the literature and practice through the development of several key policy recommendations for transportation agencies. These recommendations are designed to guide agencies in incorporating and enhancing their equity integration approaches in their long-range planning practices. The following list of bullet points provides a summary of the major policy recommendations.

- Vertical equity analyses and community participation methods should be incorporated into the scenario planning process to fully comprehend how policies affect various demographic groups, especially underserved populations.
- When assessing and selecting policies, robustness analyses and equity-centered policy development should be conducted across all scenarios.
- During the planning stage, it is important to consider how emerging mobility services and supporting technology might affect equity. An equity planning mindset for different possible scenarios is essential for future improvements.

While we found a number of key equity gaps in scenario planning, this research and its associated recommendations offer an important steppingstone for upcoming work that can help shape a transportation-for-all future.

Author contributions

The authors confirm their contribution to the paper as follows: study conception and design: Meiyu (Melrose) Pan, Alyssa Ryan, Stephen Wong, Francis Tainter, and Steve Woelfel; data collection: Meiyu (Melrose) Pan; analysis and interpretation of results: Meiyu (Melrose) Pan, Stephen Wong, Alyssa Ryan; draft manuscript preparation: Meiyu (Melrose) Pan, Alyssa Ryan, Stephen Wong; revision and review: Meiyu (Melrose) Pan, Alyssa Ryan, Stephen Wong, Francis Tainter, and Steve Woelfel. All authors approved the final version of the manuscript.

References

Aimen, David, and Anne Morris. 2012. “Practical Approaches for Involving Traditionally Underserved Populations in Transportation Decisionmaking.” *World Transit Research*.

Ange, Katharine, Kathleen Rooney, Edward A. Mierzejewski, Caroline Dwyer, Renaissance Planning, and Eno Center for Transportation. 2017. *Next Generation Scenario Planning: A Transportation Practitioner's Guide*.

Anon. 2007. *Scenario Planning Peer Workshop: Monterey, California June 2, 2006*.

Avin, Uri. 2016. *Sketch Planning Tools for Regional Sustainability*. NCHRP 08-36/TASK 117. Transportation Research Board.

Bartholomew, Keith, Reid Ewing, Gail Meakins, and University of Utah. Metropolitan Research Center. 2010. *Integrated Transportation Scenario Planning.* FHWA-HEP-10-034.

Basso, Leonardo J., and Sergio R. Jara-Díaz. 2012. "Integrating Congestion Pricing, Transit Subsidies and Mode Choice." *Transportation Research Part A: Policy and Practice* 46(6):890–900. doi: 10.1016/j.tra.2012.02.013.

Biehl, Alec, Ying Chen, Karla Sanabria-Véaz, David Uttal, and Amanda Stathopoulos. 2019. "Where Does Active Travel Fit within Local Community Narratives of Mobility Space and Place?" *Transportation Research Part A: Policy and Practice* 123:269–87. doi: 10.1016/j.tra.2018.10.023.

Bills, Tierra. 2013. "Enhancing Transportation Equity Analysis for Long Range Planning and Decision Making." UC Berkeley.

Bills, Tierra, and Joan Walker. 2017. "Looking beyond the Mean for Equity Analysis: Examining Distributional Impacts of Transportation Improvements." *Transport Policy* 54:61–69. doi: 10.1016/j.tranpol.2016.08.003.

Boisjoly, Geneviève, and Ahmed El-Geneidy. 2021. "Chapter 14 - Public Transport Equity Outcomes through the Lens of Urban Form." Pp. 223–41 in *Urban Form and Accessibility*, edited by C. Mulley and J. D. Nelson. Elsevier.

Bouaoun, Liacine, Mohamed Mouloud Haddak, and Emmanuelle Amoros. 2015. "Road Crash Fatality Rates in France: A Comparison of Road User Types, Taking Account of Travel Practices." *Accident Analysis & Prevention* 75:217–25. doi: 10.1016/j.aap.2014.10.025.

Brenman, Marc. 2007. "Transportation Inequity in the United States - A Historical Overview." *Human Rights* 34(3):7–25.

Brodie, Stefanie R. 2015. "Equity Considerations for Long-Range Transportation Planning and Program Development."

Bullard, Robert D. 2000. *Dumping In Dixie: Race, Class, And Environmental Quality*. 3rd ed. New York: Routledge.

Bullard, Robert D. 2007. *Growing Smarter: Achieving Livable Communities, Environmental Justice, and Regional Equity*. MIT Press.

Cabello, Maxwell, Michael Hyland, and Nicholas Marantz. 2023. “From State of the Practice to State of the Art: Improving Equity Analysis in Regional Transportation Plans.”

Capano, Giliberto, and Jun Jie Woo. 2018. “Designing Policy Robustness: Outputs and Processes.” *Policy and Society* 37(4):422–40. doi: 10.1080/14494035.2018.1504494.

Carlton, Gregory, and Selima Sultana. 2022. “Transport Equity Considerations in Electric Vehicle Charging Research: A Scoping Review.” *Transport Reviews* 0(0):1–26. doi: 10.1080/01441647.2022.2109775.

Chakraborty, Arnab. 2011. “Enhancing the Role of Participatory Scenario Planning Processes: Lessons from Reality Check Exercises.” *Futures* 43(4):387–99. doi: 10.1016/j.futures.2011.01.004.

Chakraborty, Arnab, Nikhil Kaza, Gerrit-Jan Knaap, and Brian Deal. 2011. “Robust Plans and Contingent Plans.” *Journal of the American Planning Association* 77(3):251–66. doi: 10.1080/01944363.2011.582394.

Chakraborty, Arnab, and Andrew McMillan. 2015. “Scenario Planning for Urban Planners: Toward a Practitioner’s Guide.” *Journal of the American Planning Association* 81(1):18–29. doi: 10.1080/01944363.2015.1038576.

Chen, Zhiwei, Yujie Guo, Amy L. Stuart, Yu Zhang, and Xiaopeng Li. 2019. “Exploring the Equity Performance of Bike-Sharing Systems with Disaggregated Data: A Story of Southern Tampa.” *Transportation Research Part A: Policy and Practice* 130:529–45. doi: 10.1016/j.tra.2019.09.048.

Chermack, Thomas J. 2011. *Scenario Planning in Organizations: How to Create, Use, and Assess Scenarios*. Berrett-Koehler Publishers.

Chittenden County Metropolitan Planning Organization and Federal Highway Administration Vermont division. 2008. *Scenario Planning Peer Workshop: Burlington, Vermont, November 9, 2007*.

Clark, Connor, Gyan P. Nyaupane, Dallen J. Timothy, and Christine Buzinde. 2022. “Scenario Planning as a Tool to Manage Tourism Uncertainties during the Era of COVID-19: A Case Study of Arizona, USA.” *Current Issues in Tourism* 25(7):1063–73. doi: 10.1080/13683500.2022.2032617.

Denver Regional Council of Governments. 2013. *Federal Highway Administration and Federal Transit Administration Scenario Planning Workshop: Denver, Colorade June 7 - 8, 2012.*

Federal Highway Administration. 2006. *Monterey, CA, Scenario Planning Peer Workshop Report*.

Federal Highway Administration. 2013. *Performance Based Planning and Programming Guidebook*.

Federal Highway Administration. 2015. *Advancing Transportation Systems Management and Operations Through Scenario Planning*.

Foster, Sheila R., and Luke W. Cole. 2000. *From the Ground Up: Environmental Racism and the Rise of the Environmental Justice Movement*. New York: NYU Press.

Garrick, Norman. 2019. *What Do We Want from Autonomous Vehicles (AVs)? Using Participatory Planning and Scenario Analysis of Alternative Features to Identify Stakeholders' Desired Outcomes from the Strategic Deployment of Emerging Transportation Technology*. 2018 Project 12.

Golub, Aaron, and Karel Martens. 2014. "Using Principles of Justice to Assess the Modal Equity of Regional Transportation Plans." *Journal of Transport Geography* 41:10–20. doi: 10.1016/j.jtrangeo.2014.07.014.

Guo, Yujie, Zhiwei Chen, Amy Stuart, Xiaopeng Li, and Yu Zhang. 2020. "A Systematic Overview of Transportation Equity in Terms of Accessibility, Traffic Emissions, and Safety Outcomes: From Conventional to Emerging Technologies." *Transportation Research Interdisciplinary Perspectives* 4:100091. doi: 10.1016/j.trip.2020.100091.

Guthrie, Andrew, Yingling Fan, and Kirti Vardhan Das. 2017. "Accessibility Scenario Analysis of a Hypothetical Future Transit Network: Social Equity Implications of a General Transit Feed Specification–Based Sketch Planning Tool." *Transportation Research Record* 2671(1):1–9. doi: 10.3141/2671-01.

Guzman, Luis A., and Philipp Hessel. 2022. "The Effects of Public Transport Subsidies for Lower-Income Users on Public Transport Use: A Quasi-Experimental Study." *Transport Policy* 126:215–24. doi: 10.1016/j.tranpol.2022.07.016.

Higgins, Nathan. 2019. *GUIDANCE FOR SCENARIO PLANNING*. NCHRP 08-36/Task 145. Cambridge Systematics, Inc.

Houston, Douglas, Jun Wu, Paul Ong, and Arthur Winer. 2004. "Structural Disparities of Urban Traffic in Southern California: Implications for Vehicle-Related Air Pollution Exposure in Minority and High-Poverty Neighborhoods." *Journal of Urban Affairs* 26(5):565–92. doi: 10.1111/j.0735-2166.2004.00215.x.

Hughes-Cromwick, MacPherson. 2019. "Supporting Late-Shift Workers: Their Transportation Needs and the Economy." *American Public Transportation Association*. Retrieved December 5, 2022 (https://www.apta.com/research-technical-resources/research-reports/supporting-late-shift-workers-their-transportation-needs-and-the-economy/).

John A. Volpe National Transportation Systems. 2010. *New Trends in Transportation and Land Use Scenario Planning: Five Case Studies of Regional and Local Scenario Planning Efforts*.

John A. Volpe National Transportation Systems Center, ed. 2011. *Federal Highway Administration Scenario Planning Guidebook*.

Karner, Alex, Jonathan London, Dana Rowangould, and Kevin Manaugh. 2020. “From Transportation Equity to Transportation Justice: Within, through, and beyond the State.” *Journal of Planning Literature* 35(4):440–59.

Kaufman, Matthew, Sonika Sethi, Ellen Partridge, Rudy Faust, and Leidos. 2020. *Mobility on Demand Business Models Scenario Planning Template*. FHWA-JPO-20-797.

LA Metro. 2019. “UNDERSTANDING HOW WOMEN TRAVEL.”

Lee, Richard, Ipek Sener, and Nathan Jones. 2017. “Understanding the Role of Equity in Active Transportation Planning in the United States.” *Transport Reviews* 37(2):211–26. doi: 10.1080/01441647.2016.1239660.

Lee, Stephanie, Mike Tremble, Justine Vaivai, Gregory Rowangould, Mohammad Tayarani, Amir Poorfakhraei, Inc. Ecosystem Management, and University of New Mexico. 2015. *Central New Mexico Climate Change Scenario Planning Project: Final Report*.

Lempert, Robert, James Syme, George Mazur, Debra Knopman, Garett Ballard-Rosa, Kacey Lizon, and Ifeanyi Edochie. 2020. “Meeting Climate, Mobility, and Equity Goals in Transportation Planning Under Wide-Ranging Scenarios.” *Journal of the American Planning Association*.

Lewis, Elyse O’Callaghan, Don MacKenzie, and Jessica Kaminsky. 2021. “Exploring Equity: How Equity Norms Have Been Applied Implicitly and Explicitly in Transportation Research and Practice.” *Transportation Research Interdisciplinary Perspectives* 9:100332. doi: 10.1016/j.trip.2021.100332.

Linovski, Orly, Dwayne Marshall Baker, and Kevin Manaugh. 2018. “Equity in Practice? Evaluations of Equity in Planning for Bus Rapid Transit.” *Transportation Research Part A: Policy and Practice* 113:75–87. doi: 10.1016/j.tra.2018.03.030.

Litman, Todd. 2018. “Evaluating Transportation Equity.” *World Transport Policy & Practice* 8:50–65.

Litman, Todd. 2022. *Evaluating Transportation Equity Guidance for Incorporating Distributional Impacts in Transport Planning*. Victoria Transport Policy Institute.

Liu, Yuandong, Ho-Ling Hwang, Majbah Uddin, Tim Reuscher, and Shih-Miao Chin. 2022. *Travel Patterns and Characteristics of Elderly Population in New York State: 2017 Update*. ORNL/TM-2022/2614. Oak Ridge National Lab. (ORNL), Oak Ridge, TN (United States). doi: 10.2172/1899831.

Luo, Xiaoling, Xiaobo Ma, Matthew Munden, Yao-Jan Wu, and Yangsheng Jiang. 2022. “A Multisource Data Approach for Estimating Vehicle Queue Length at Metered On-Ramps.” *Journal of Transportation Engineering, Part A: Systems* 148(2):04021117. doi: 10.1061/JTEPBS.0000622.

Lyons, Glenn, Charlene Rohr, Annette Smith, Anna Rothnie, and Andrew Curry. 2021. "Scenario Planning for Transport Practitioners." *Transportation Research Interdisciplinary Perspectives* 11:100438. doi: 10.1016/j.trip.2021.100438.

McBrien, Rachel Strauss and John A. Volpe National Transportation Systems Center (U.S.). 2018. *Wichita Area Metropolitan Planning Organization Scenario Planning Workshop: Sponsored by the Federal Highway Administration*. FHWA-HEP-19-018.

Michigan Dept. of Transportation and Tri-county Regional Planning Commission. 2010. *Federal Highway Administration Scenario Planning: Peer Workshop, East Lansing, Michigan July 13, 2010*.

Moore, Sarah S., Nathaniel E. Seavy, Matt Gerhart, and California Coastal Conservancy. 2013. *Scenario Planning for Climate Change Adaptation: A Guidance for Resource Managers*. United States of America - gov.

Neiner, Jennifer A., Elizabeth H. Howze, and Mary L. Greaney. 2004. "Using Scenario Planning in Public Health: Anticipating Alternative Futures." *Health Promotion Practice* 5(1):69–79. doi: 10.1177/1524839903257692.

Page, Matthew J., Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, Roger Chou, Julie Glanville, Jeremy M. Grimshaw, Asbjørn Hróbjartsson, Manoj M. Lalu, Tianjing Li, Elizabeth W. Loder, Evan Mayo-Wilson, Steve McDonald, Luke A. McGuinness, Lesley A. Stewart, James Thomas, Andrea C. Tricco, Vivian A. Welch, Penny Whiting, and David Moher. 2021. "The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews." *International Journal of Surgery* 88:105906. doi: 10.1016/j.ijsu.2021.105906.

Page, Stephen John, Ian Yeoman, Joanne Connell, and Chris Greenwood. 2010. "Scenario Planning as a Tool to Understand Uncertainty in Tourism: The Example of Transport and Tourism in Scotland in 2025." *Current Issues in Tourism*. doi: 10.1080/13683500802613519.

Pan, Meiyu (Melrose), and Alyssa Ryan. 2023. "Equity in Active and Shared Transportation: An Investigation of Barriers and Individual and Contextual Factors." *Transportation Research Record* 03611981221143106. doi: 10.1177/03611981221143106.

Pandey, Bhartendu, Christa Brelsford, and Karen C. Seto. 2022. "Infrastructure Inequality Is a Characteristic of Urbanization." *Proceedings of the National Academy of Sciences* 119(15):e2119890119. doi: 10.1073/pnas.2119890119.

Park, Yongha, Jerry Mount, Luyu Liu, Ningchuan Xiao, and Harvey J. Miller. 2020. "Assessing Public Transit Performance Using Real-Time Data: Spatiotemporal Patterns of Bus Operation Delays in Columbus, Ohio, USA." *International Journal of Geographical Information Science* 34(2):367–92. doi: 10.1080/13658816.2019.1608997.

Peregrino, Ana Alice, Soham Pradhan, Zhicheng Liu, Nivan Ferreira, and Fabio Miranda. 2021. "Transportation Scenario Planning with Graph Neural Networks."

Pereira, Rafael HM, and Alex Karner. 2021. "Transportation Equity." *International Encyclopedia of Transportation* 1:271–77.

Ravetz, Joe, Aleksi Neuvonen, and Raine Mäntysalo. 2021. "The New Normative: Synergistic Scenario Planning for Carbon-Neutral Cities and Regions." *Regional Studies* 55(1):150–63. doi: 10.1080/00343404.2020.1813881.

Renzi, Adriano Bernardo, and Sydney Freitas. 2015. "The Delphi Method for Future Scenarios Construction." *Procedia Manufacturing* 3:5785–91. doi: 10.1016/j.promfg.2015.07.826.

Rohr, Charlene, Liisa Ecola, Johanna Zmud, Fay Dunkerley, James Black, and Eleanor Baker. 2016. *Travel in Britain in 2035: Future Scenarios and Their Implications for Technology Innovation*. RAND Corporation.

Rowangould, Dana, Alex Karner, and Jonathan London. 2016. "Identifying Environmental Justice Communities for Transportation Analysis." *Transportation Research Part A: Policy and Practice* 88:151–62. doi: 10.1016/j.tra.2016.04.002.

Sadatsafavi, Hessam, Amy A. Kim, Stuart D. Anderson, and Peter Bishop. 2019. "Scenario Planning Application in US Highway Transportation Industry." *Journal of Infrastructure Systems* 25(2):05019002. doi: 10.1061/(ASCE)IS.1943-555X.0000474.

Sanchez, Thomas. 2019. *The Right to Transportation: Moving to Equity*. New York: Routledge.

Sanchez, Thomas, Rich Stolz, and Jacinta Ma. 2003. "Moving to Equity: Addressing Inequitable Effects of Transportation Policies on Minorities."

Shaheen, Susan A., Madonna L. Camel, and Kunik Lee. 2013. "U.S. Integrated Transportation Systems in the Future, 2030 to 2050: Application of a Scenario Planning Tool." *Transportation Research Record* 2380(1):99–107. doi: 10.3141/2380-11.

Shaheen, Susan, Corwin Bell, Adam Cohen, Balaji Yelchuru, and Inc. Booz Allen Hamilton. 2017. *Travel Behavior: Shared Mobility and Transportation Equity*. PL-18-007.

Shaheen, Susan, Madonna Camel, and Kunik Lee. 2013. "Exploring the Future of Integrated Transportation Systems in the United States from 2030 to 2050: Application of a Scenario Planning Tool."

Shaheen, Susan, and Stephen Wong. 2021. "Future of Public Transit and Shared Mobility: Scenario Planning for COVID-19 Recovery." doi: 10.7922/G2NC5ZGR.

Shaheen, Susan, and Stephen Wong. 2023. "The Future of Public Transit and Shared Mobility: Policy Actions and Research Options for COVID-19 Recovery." Pp. 313–31 in *Pandemic in the Metropolis: Transportation Impacts and Recovery*, *Springer Tracts on*

Transportation and Traffic, edited by A. Loukaitou-Sideris, A. M. Bayen, G. Circella, and R. Jayakrishnan. Cham: Springer International Publishing.

Sheldon, Tamara L. 2022. "Evaluating Electric Vehicle Policy Effectiveness and Equity." *Annual Review of Resource Economics* 14(1):669–88. doi: 10.1146/annurev-resource-111820-022834.

Snyder, Kenneth. 2017. "Advancement of Performance-Based Scenario Planning for Regional Planning and Decision-Making: A Synthesis Report." *American Planning Association*. Retrieved December 6, 2022 (https://www.planning.org/knowledgebase/resource/9146914/).

Strauss, Rachel. 2016. *New York Metropolitan Transportation Council Scenario Planning Workshop : Sponsored by the Federal Highway Administration*. edited by John A. Volpe National Transportation Systems Center (U.S.).

Transportation Research Board. 2022. "TRB Scenario Planning Conference." *TRB Scenario Planning Conference Session Schedule*. Retrieved December 7, 2022 (https://trb.secure-platform.com/a/solicitations/85/sessiongallery/schedule?dayId=171&searchParams=%7B%22pageIndex%22%3A0,%22sortMode%22%3A%22SessionStartTime%22,%22sortDirection%22%3A%22Ascending%22,%22sortByFieldId%22%3Anull,%22displayMode%22%3Anull,%22filterByFieldValues%22%3A%5B%5D,%22filterByTextValue%22%3Anull,%22filterByFavorites%22%3Afalse,%22filterByScheduleRoomIds%22%3A%5B%5D,%22filterBySessionTypeIds%22%3A%5B%5D,%22filterByScheduleDayIds%22%3A%5B%5D,%22filterByScheduleTimeSlotIds%22%3A%5B%5D,%22isScheduleOtherEventSearchAllowed%22%3Atrue%7D).

Tu, Chelsea, and Richard Marcantonio. 2016. "A FRAMEWORK FOR EQUITABLE INVESTMENT." *Berkeley Planning Journal* 28(1). doi: 10.5070/BP328133861.

Twaddell, Hannah, Alanna McKeeman, Michael Grant, Jessica Klion, Uri Avin, Kate Ange, and Mike Callahan. 2016. "Supporting Performance-Based Planning and Programming through Scenario Planning."

US Department of Transportation. 2022. *Equity Action Plan*.

Walker, Warren E., S. Adnan Rahman, and Jonathan Cave. 2001. "Adaptive Policies, Policy Analysis, and Policy-Making." *European Journal of Operational Research* 128(2):282–89. doi: 10.1016/S0377-2217(00)00071-0.

Wei, Ran, Xiaoyue Liu, Yongjian Mu, Liming Wang, Aaron Golub, and Steven Farber. 2017. "Evaluating Public Transit Services for Operational Efficiency and Access Equity." *Journal of Transport Geography* 65:70–79. doi: 10.1016/j.jtrangeo.2017.10.010.

Zegras, Christopher, Joseph Sussman, and Christopher Conklin. 2004. "Scenario Planning for Strategic Regional Transportation Planning." *Journal of Urban Planning and Development* 130(1):2–13. doi: 10.1061/(ASCE)0733-9488(2004)130:1(2).